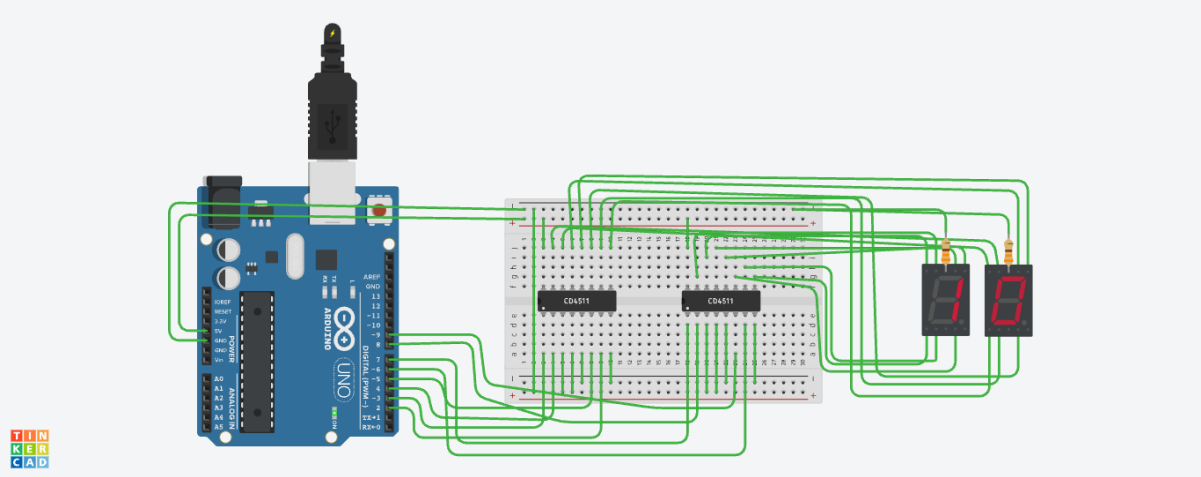
0–30 Counter Using Two 7-Segment Displays

Circuit Diagram

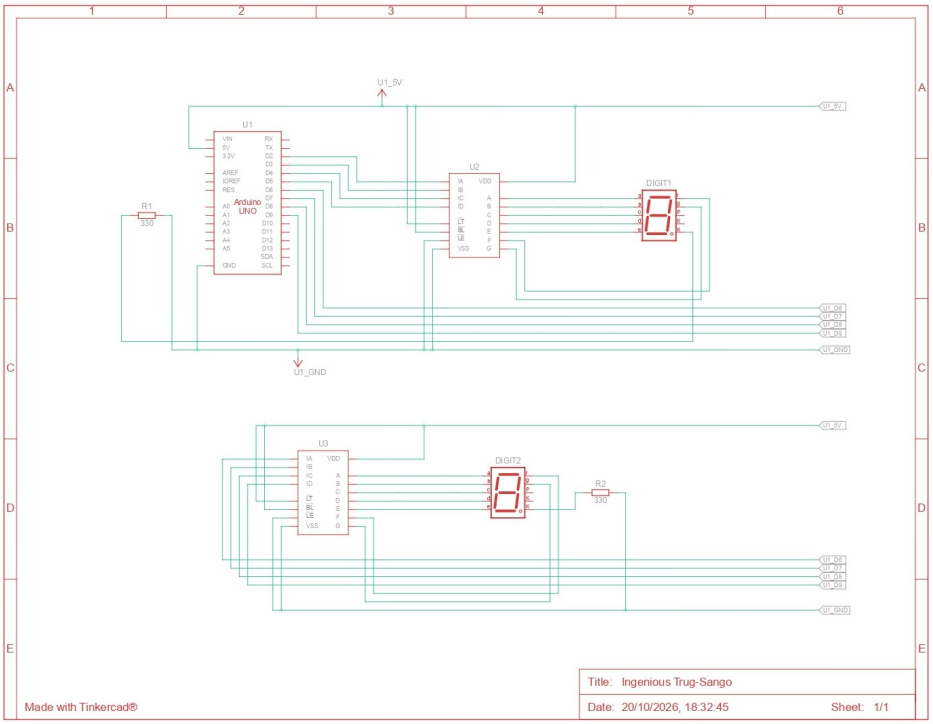


Component List

Component List — Download CSV

Name	Quantity	Component
U1	1	Arduino Uno R3
3	1	Breadboard Small
U2 U3	2	7-Segment Decoder
Digit1 Digit2	2	Cathode 7 Segment Display
R1 R2	2	330 Ω Resistor

Schematic


1
2
3
4
5
6
A
B
C
D
E
U1_5V
U1
Arduino UNO
R1
330
U2
DIGIT1
U1_GND
U3
DIGIT2
R2
330
Made with Tinkercad®
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Date: 20/10/2026, 18:32:45
Sheet: 1/1

Code

```
// Arduino pins connected to the BCD inputs of each decoder
// Index 0 = bit A (LSB) ... index 3 = bit D (MSB)
const int unitsPins[4] = {2, 3, 4, 5};
const int tensPins[4]  = {6, 7, 8, 9};

const unsigned long STEP_MS = 1000;   // time each number stays on the display
const int MAX_COUNT = 30;             // counter runs from 0 to MAX_COUNT

int count = 0;

void setup() {
  for (int i = 0; i < 4; i++) {
    pinMode(unitsPins[i], OUTPUT);
    pinMode(tensPins[i], OUTPUT);
  }
  showNumber(count);                  // show 00 at power-on
}

// Writes one decimal digit (0-9) as 4-bit BCD on the given pins
void writeBCD(const int pins[], int digit) {
  for (int i = 0; i < 4; i++) {
    digitalWrite(pins[i], (digit >> i) & 0x01);
  }
}

// Splits a number into tens and units and sends each to its decoder
void showNumber(int n) {
  int tens  = n / 10;
  int units = n % 10;
  writeBCD(tensPins, tens);
  writeBCD(unitsPins, units);
}

void loop() {
  showNumber(count);
  delay(STEP_MS);

  count++;
  if (count > MAX_COUNT) {
    count = 0;                        // wrap around after 30
  }
}
```